

U1-CO2 PROBLEM

1. **PROBLEM STATEMENT:** A hospital has to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) by satisfying these constraints:
 - D1 cannot work Night.
 - D2 must be scheduled before D3.
 - Only one doctor is allowed in each shift.
 - D3 cannot work Morning.
 - Every doctor must be assigned exactly one shift.

2. **PROBLEM STATEMENT:** A robot moves in a 5×5 grid from Start (S) to Goal (G). The robot can move Up, Down, Left, and Right, each move costs 1, it cannot pass through obstacles, and it uses the Manhattan Distance heuristic to find the shortest path.

3. **PROBLEM STATEMENT:** An autonomous rescue robot is deployed in a disaster-affected building to rescue survivors. The robot starts from S and must reach the survivor at G. It can move Up, Down, Left, and Right, cannot pass through obstacles, has limited sensing, and must minimize the total travel cost while updating its knowledge dynamically.